

Cover Letter

Original submission to IEEE Transactions on Signal Processing

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To: The Editor-in-Chief, *IEEE Transactions on Signal Processing*

From: Bin Nong, Weihong Fu, and Zhuoyun Jiang

Corresponding email: nongbin@tfsfwufe.edu.cn

Manuscript title: Structure, Not Loss: Sync-Aware Joint Detection for Semi-Blind Single-Channel Co-Frequency Reception

Manuscript type: Regular Paper

Dear Editor,

We are pleased to submit the above manuscript for consideration in *IEEE Transactions on Signal Processing*. The paper addresses a gap in the single-channel blind source separation (SC-BSS) literature: the field trains and ranks separators by waveform fidelity (SI-SDR/SDR), but a receiver's deliverable is bits, and we show that for co-frequency communication mixtures the two decouple *structurally*. We give the decoupling a theory backbone, locate its mechanism experimentally, and derive the receiver structure the analysis points to. The work combines signal-processing theory (detection error floors, identifiability, Fisher analysis) with algorithm design and statistically controlled evaluation, and we believe it fits the Transactions' scope in signal processing theory, methods, and communications applications.

Key contributions.

1. **Theory with simulation validation.** For the symbol-rate two-source mixture model we prove a joint-MAP union bound governed by the sum-constellation minimum distance $d_{\min}(\varphi)$ and an *exact* separate-detection error floor under a discrete interferer (e.g. the 1/2 high-SNR floor for QPSK+QPSK at 0 dB SIR); a finite-symmetry identifiability proposition showing that only the effective per-source complex coefficients are recoverable from the population mixture law, up to the constellations' rotational group, and only through the joint discrete structure; and an exact two-tone frequency-CRB inflation analysis (median $1.7\times$ under the benchmark's offset law) contrasted with a communication-waveform Fisher analysis showing no two-source coupling once symbols are known. Each statement is accompanied by a Monte-Carlo or measurement validation, including a direct estimator-versus-CRB experiment at $K=2$.
2. **Architecture diagnosis.** A controlled experiment ladder (E3–E4) shows that per-slot blind synchronisation is captured by the residual interferer's spectral line (SNR-flat +8–11 SER points, reproduced cross-architecture with a $26\times$ larger published separator), that slot-aided detection loses to mixture-level detection because mask separation distorts the linear observation model, and that the separate-detection floor is a property of the linear-residual regime rather than of separation as such.
3. **A training-free joint receiver.** The ISI-aware ECM joint synchroniser-detector on the raw $K=2$ mixture — no oracle carriers, phases, or symbols during inference — ties the strongest waveform route at the paper's separator quality (pooled SER 0.531 vs. 0.534), wins at matched oracle frequencies against even the stronger CNSE-based route (0.411 vs. 0.444), and leads every waveform route under genie-free differential scoring (0.372 vs. 0.438). A learning-free blind carrier synchroniser matches an oracle receiver within 0.5 SER points for BPSK/QPSK bursts at $K=1$.

4. **Honest envelope.** The operating boundaries are measured, not assumed: blind modulation-pair selection fails at the classifier, the tested $K=3$ acquisition strategy fails by phantom-pair lock, and frequency-separation, timing, and amplitude-imbalance robustness are swept explicitly.

Statistical standard. The unit of inference is the mixture burst. Headline contrasts are paired per-burst (700 paired bursts on the reference grid; 3500 across five independent test grids) with paired-bootstrap confidence intervals and Wilcoxon signed-rank tests, and ties are claimed only when the paired interval contains zero. A demodulation-aware loss-level control and a strong published SC-BSS baseline (CNSE) are included as controls.

Reproducibility. All code — the synthetic generator, the baseline separator and its training script, the receiver modules, and every evaluation script behind the tables — is publicly available at https://github.com/BinNong/single_channel_source_seperation/tree/main/paper6_sync_jd; the benchmark is fully synthetic and reproducible from the stated seeds (deterministic test seed 99999), and the exact command behind every number is recorded in the repository’s experiment log.

Related submissions. Companion manuscripts on closed-set and open-world separator design are under review elsewhere; they share no results with the present paper, which is fully self-contained (models, generator, and baselines are re-derived or vendored, and only published work is cited).

Originality and ethics.

- The manuscript is original, has not been published elsewhere, and is not under consideration by any other journal.
- All authors have approved the submission; the authors have no conflicts of interest to declare.
- No human subjects or animal experiments are involved; all experiments use synthetic signals from our released generator.

Author contributions. B. Nong conceived the study, derived the theory, implemented the receiver and experiments, and drafted the manuscript. W. Fu contributed to the conceptualisation of the evaluation framework. Z. Jiang contributed resources and validation. All authors reviewed and approved the final manuscript.

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Suggested reviewers. We have no specific reviewer suggestions; we trust the editorial board to identify expert reviewers in statistical signal processing, co-frequency interference management, and synchronisation and detection theory.

Thank you for considering our manuscript. We look forward to hearing from you.

Sincerely,

Bin Nong

Tianfu College of Southwestern University of Finance and Economics, Mianyang, China
`nongbin@tfswufe.edu.cn`

Weihong Fu

School of Telecommunications Engineering, Xidian University, Xi'an, Shaanxi, China

Zhuoyun Jiang

School of Health Science and Engineering, University of Shanghai for Science and Technology,
Shanghai, China